momentum at the same time. But because quantum physics doesn't let you simultaneously predict the position and momentum of a single particle, EPR argued that it must be *incomplete*—there must be features of the world that quantum physics doesn't account for. The EPR paper closes by holding out hope for a better theory that can account for those things: "While we have thus shown that the wave function does not provide a complete description of the physical reality, we left open the question of whether or not such a description exists. We believe, however, that such a theory is possible."

The most famous scientist in the world blasting away at a well-known (if not well-understood) theory in such harsh terms made for a media frenzy, of course—especially when Podolsky leaked the story to the press early. "EINSTEIN ATTACKS QUANTUM THEORY" blared the *New York Times* on May 4, 1935, several days before the EPR paper was published. "Scientist and Two Colleagues Find It Is Not 'Complete' Even Though 'Correct.'" Einstein, furious, sent a statement to the newspaper in reply: "Any information upon which the article 'Einstein Attacks Quantum Theory'… is based was given to you without my authority. It is my invariable practice to discuss scientific matters only in the appropriate forum and I deprecate advance publication of any announcement in regard to such matters in the secular press."

Podolsky's leak wasn't the only reason Einstein was upset. Despite the fact that his name was on the EPR paper, Einstein hadn't actually written it himself—and he wasn't happy with it either. Shortly after its publication, Einstein told Schrödinger that the EPR paper "was written by Podolsky after much discussion. Still, it did not come out as well as I had originally wanted; rather the essential thing was, so to speak, smothered by the [mathematics]." Later in the same letter, he said that he "couldn't care less" about the uncertainty principle; his real problem with quantum physics had nothing to do with that.

For Einstein, the crucial bit of the EPR thought experiment once again had to do with locality. If you measure A's momentum, you know B's momentum too. But because B is far away from A, then, assuming locality, there's no way that making a measurement on A could have affected B immediately. B's momentum must have been set when A and B collided, just like billiard balls.